

EXACT 2026

Physics API

Single /predict endpoint

Endpoint

The public URL points to /predict and returns a JSON list with one result object per query.

Type 1 route

Premise-based questions are converted to a shared logic form and checked with Z3. The answer is Yes, No, Uncertain, or an MCQ option.

Type 2 route

Numerical physics questions are solved with deterministic formula code or with a parser plus calculator fallback.

Submitted system

The API routes each request by its official type field. Type 1 uses formal verification over supplied premises. Type 2 computes physical quantities and units using formula solvers, parser output, and calculator checks.

System Components

Component	Role
Type 1 logic	Converts premises and target statements into one logic vocabulary, then verifies entailment with Z3.
Type 2 numeric	Solves common physics patterns directly and uses parser plus calculator fallback for unfamiliar wording.
Calculator	Owns arithmetic, formula application, and unit conversion.
Explanation	Explains locked results without changing answer, unit, or calculation.

Response contract

Every /predict response is a JSON list. Each item contains query_id, answer, unit, explanation, premises_used, and reasoning. Debug fields are returned only when requested.

Runtime Flow

- **Receive**

Accept a JSON object or list of objects with `query_id`, `type`, `query`, `premises`, and `options`.

- **Route**

`type1` with `premises` goes to the logic verifier. `type2` goes to the numeric physics path.

- **Solve**

Type 1 checks entailment. Type 2 computes the requested quantity with formulas, parser output, and calculator validation.

- **Verify**

The API checks output shape, units, confidence, and whether a safer fallback route is needed.

- **Return**

Each result includes `answer`, `unit`, `explanation`, `premises_used`, and `reasoning`.

Key guardrail

The model-facing layer is not trusted to own final arithmetic. For Type 2, numbers and units are recomputed after extraction. For Type 1, the final label comes from a formal entailment check whenever premises are supplied.

Reliability Design

Type 1 is formalized

The language model proposes a parser-safe logic representation. Z3 decides whether the premises entail the target statement or an answer option.

Type 2 is calculated

The language model may normalize wording, but final values come from deterministic formula code, unit conversion, and calculator validation.

Check	Purpose
Input contract	Return a JSON list with one result object per query.
Logic consistency	For Type 1, compare the converted target against converted premises using Z3.
Quantity grounding	For Type 2, extracted numbers and units must be grounded in the question.
Unit consistency	Values are converted before final formatting or claim comparison.

Deployment Summary

Kaggle runtime

Upload `physics_api_dataset.zip` to Kaggle, add it as notebook input, enable GPU, set `NGROK_AUTHTOKEN`, and run the deploy notebook cells in order. The notebook prints the final `prediction_url` ending in `/predict`.

Required public URL

Submit only the `/predict` URL printed by the notebook. The API must stay alive during the grading slot.

Returned format

The response is always a JSON list. Type 2 includes answer and unit. Type 1 returns the selected label and explanation.

- **Solution PDF**

This document. Replace later only if you add a separate Type 1 write-up.

- **Source code ZIP**

Generated without README, cache, preview, or test files.

- **URLs file**

Update after ngrok prints the final `/predict` URL.

- **Notation mapping CSV**

Included as a required portal file.